The Strong Vector-Valued Hopf–Dunford–Schwartz Inequality: A Later-Literature Answer to Part (2) of arXiv:1202.5838

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Status

Literature already answered (affirmative), with limited scope. Theorem 1 and Corollary 3 of Xu answer part (2) of the two-part conjecture of Charpentier–Deleaval. The weak endpoint in part (1) is not claimed here.

Source conjecture

Let $(T_t)_{t\geq 0}$ be a strongly continuous semigroup whose members are contractions on both $L^1(\Omega)$ and $L^\infty(\Omega)$, and define

$$M_T f = \sup_{a>0} \frac{1}{a} \left| \int_0^a T_t f \, dt \right|, \qquad \overline{M}_T (f_j)_j = (M_T f_j)_j.$$

On PDF page 5 of arXiv:1202.5838, Charpentier–Deleaval conjecture, for every $1 < q < \infty$, both a weak $L^1(\ell^q)$ endpoint and the following part (2): for every $1 < p < \infty$,

$$\|\overline{M}_T f\|_{L^p(\ell^q)} \lesssim_{p,q} \|f\|_{L^p(\ell^q)}.$$

Explicit answering theorem

Xu explicitly identifies the question on PDF page 1 of arXiv:1402.2344: after citing Charpentier–Deleaval, he says that the missing strong range is answered affirmatively. His Theorem 1 on PDF page 2 states that if $1 < p < \infty$, (T_t) is a strongly continuous semigroup of regular contractions on $L^p(X)$, and E is any UMD Banach lattice, then

$$\|M(T)f\|_{L^p(X;E)} \lesssim_{p,E} \|f\|_{L^p(X;E)}.$$

Corollary 3 gives this conclusion for semigroups of contractions on $L^r(X)$ for every $1 \leq r \leq \infty$ (with strong continuity on L^2).

Taking $E = \ell^q$, which is a UMD Banach lattice for every $1 < q < \infty$, gives exactly the source conjecture's part (2), simultaneously for all $1 < p, q < \infty$. Xu also notes immediately before Theorem 1 that an operator contractive on L^1 or L^∞ is regular; thus the source's contraction hypothesis is the relevant one. Because Xu cites the source and states that he answers its question, this is an explicit literature answer, not an agent-inferred theorem coincidence.

Scope limitation

This packet closes only the strong part (2). Xu's Problem 10 on PDF page 9 asks for the corresponding weak $L^1(X;E)$ estimate and states that it is open even for $E = \ell^q$, $1 < q < \infty$. A bounded 2026 audit of the source and answer citation neighborhoods found no later paper claiming to close that endpoint. The present run's dilation, stopping-time, dyadic-time, and finite-state counterexample attempts did not produce a proof or disproof, so no endpoint result is promoted.

Search evidence

The identification was checked in the official PDFs and parsed sources of arXiv:1202.5838 and arXiv:1402.2344. Exact-title and exact-endpoint searches were run in the local arXiv corpus and on arXiv, and the citation lists of both papers were audited through 2025. Later hits concern strong maximal or variation estimates, functional calculus, and special semigroups; none found in this bounded audit states a resolution of Xu's Problem 10.

References

[1] S. Charpentier and L. Deleaval, *On a vector-valued Hopf–Dunford–Schwartz lemma*, arXiv:1202.5838; Positivity 17 (2013), 899–910.

[2] Q. Xu, H^∞ *functional calculus and maximal inequalities for semigroups of contractions on vector-valued* L_p*-spaces*, arXiv:1402.2344; Int. Math. Res. Not. IMRN 2015, no. 14, 5715–5732, Theorem 1 and Corollary 3.